

Artificial intelligence

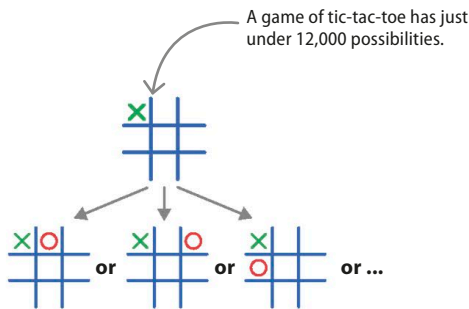
Artificial Intelligence (AI) is a family of algorithms that mimics the human behaviours of learning or reasoning. Most AIs are designed for analysing data, pattern recognition, and simulations.

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Playing with intelligence

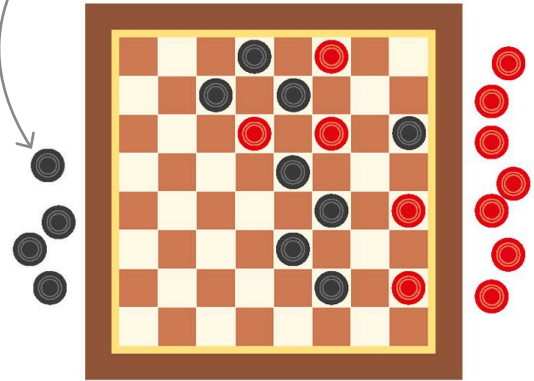
One way the intelligence of an AI can be measured is by how it makes choices based on a set of rules, such as the rules of a game. Once the rules have been outlined, the AI simulates the outcomes of the various moves open to it, and takes stock at each stage of the game to work out if it is going well.



△ Likely outcomes

The AI simulates a game, move by move, working out how a particular move is likely to work out, and how an opponent may choose to respond.

A well-chosen heuristic gives the illusion of an intelligent opponent.



△ Rules of thumb

The AI uses heuristics (also known as “rules of thumb”) to work out how the game is going. The heuristic might be how many pieces it has left in relation to the opponent.

Machine learning

Machine learning is the branch of computer science that gives a computer the ability to learn to do something with data, rather than explicitly programming it. A machine-learning algorithm learns by sifting through data bit by bit, and gradually building a model of which criteria are important and which are not for its chosen task. Eventually, it works out a way to do a task on its own based on what it knows and what it can do. Machine-learning algorithms can be used to play games, optimize transport schedules, or be the basis for robots to understand how to do work or recognize people.

▷ Gradually becoming intelligent

The more data that a machine-learning algorithm processes, the closer it gets to understanding and ultimately acting upon its task.

